

MATHEMATICAL FOUNDATIONS FOR CLUSTERING, SIMILARITY, AND NETWORK ANALYSIS

CONTENTS

Two kinds of mathematical frameworks are discussed in this chapter: clustering algorithms, and metrics for evaluating similarity and clustering performance.

1 LINEAR ALGEBRA DISTANCE CALCULATIONS

2 SIMILARITY METRICS

2.1 *Tanimoto Coefficient (also known as the Jaccard Index)*

Range: 0-1 **Higher is More Similar:** True

We define the Tanimoto Coefficient (TC) between two sets (A, B) as such:

$$TC(A, B) = TC(B, A) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

The related Jaccard Distance is simply $1 - TC$.

This is a favorite for comparing bitmaps (i.e. chemical fingerprints), in part because calculating the intersection and union of two bitvectors is very efficient.

As an example, given ligand A and ligand B with fingerprints:

$$A = |0|1|0|1|0|$$

$$B = |0|1|1|0|0|$$

$$TC(A, B) = \frac{|0|1|0|0|0|}{|0|1|1|1|0|} = \frac{1}{3} = 0.33$$

2.2 *Dice-Sørensen Coefficient*

Range: 0-1 **Higher is More Similar:** True

Following the conventions above, the Dice-Sørensen Coefficient can be calculated by:

$$DC(A, B) = \frac{2|A \cap B|}{|A| + |B|}$$

Although similar to the Tanimoto Coefficient, this function does not satisfy the triangle inequality and is much less frequently used.

2.3 Tversky Index

Range: 0-1 Higher is More Similar: True

The Tversky Index is a generalization of the Tanimoto Coefficient and Dice-Sørensen Coefficient, and is defined as:

$$\Pi(A, B) = \frac{|A \cap B|}{\alpha|A \setminus B| + \beta|B \setminus A| + |A \cap B|}$$

Note that the Tanimoto Coefficient is a special case of the Tversky Index where $\alpha = \beta = 1$, and the Dice-Sørensen Coefficient is a special case where $\alpha = \beta = 0.5$. Although α and β need not sum to 1, they are of special interest. If $\alpha \neq \beta$, the Tversky Index is asymmetric, and the value of α and β can be tuned to emphasize the importance of the first or second set.

2.3.1 Proving that the Tanimoto Coefficient is a Special Case of the Tversky Index

$$|A \setminus B| = |A| - |A \cap B|$$

$$|B \setminus A| = |B| - |A \cap B|$$

$$\text{Let } \alpha = \beta = 1$$

$$\Pi(A, B) = \frac{|A \cap B|}{|A \setminus B| + |B \setminus A| + |A \cap B|}$$

$$\Pi(A, B) = \frac{|A \cap B|}{|A| - |A \cap B| + |B| - |A \cap B| + |A \cap B|}$$

$$\Pi(A, B) = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

$$\Pi(A, B) = TC(A, B)$$

2.3.2 Proving that the Dice-Sørensen Coefficient is a Special Case of the Tversky Index

$$\text{Let } \alpha = \beta = 0.5$$

$$\Pi(A, B) = \frac{|A \cap B|}{0.5|A \setminus B| + 0.5|B \setminus A| + |A \cap B|}$$

$$\Pi(A, B) = \frac{|A \cap B|}{0.5(|A| - |A \cap B|) + 0.5(|B| - |A \cap B|) + |A \cap B|}$$

$$\Pi(A, B) = \frac{|A \cap B|}{0.5|A| - 0.5|A \cap B| + 0.5|B| - 0.5|A \cap B| + |A \cap B|}$$

$$\Pi(A, B) = \frac{2|A \cap B|}{|A| + |B|}$$

$$\Pi(A, B) = DC(A, B)$$

2.4 Cosine Similarity

3 CLUSTERING METRICS

3.0.1 Silhouette Coefficient

The silhouette coefficient is a measure of how similar an object is to its own cluster compared to other clusters. First we calculate the mean intra-cluster distance a_i by taking the average of each distance $d(i, j)$ for each object i in cluster C_i :

$$a_i = \frac{1}{|C_i| - 1} \sum_{j \in C_i, i \neq j} d(i, j)$$

Next we calculate the mean nearest-cluster distance b_i for each object i in cluster C_i :

$$b_i = \min_{j \neq i} \frac{1}{|C_j|} \sum_{j \in C_j} d(i, j)$$

Then we can define the silhouette coefficient is defined for each object i as:

$$SC_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

Finally, by taking the mean silhouette coefficient over all objects, the final value will be between -1 and 1, where a high value indicates a good clustering.